



Vedanta Bharati
Vedanta Institute of
Studies and Research



And
The Department Of Electronics And Communication Engineering
Present

HACKWARE

36 HOUR HARDWARE HACKATHON

TEAM NAME AND MEMBERS

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THEME DETAILS

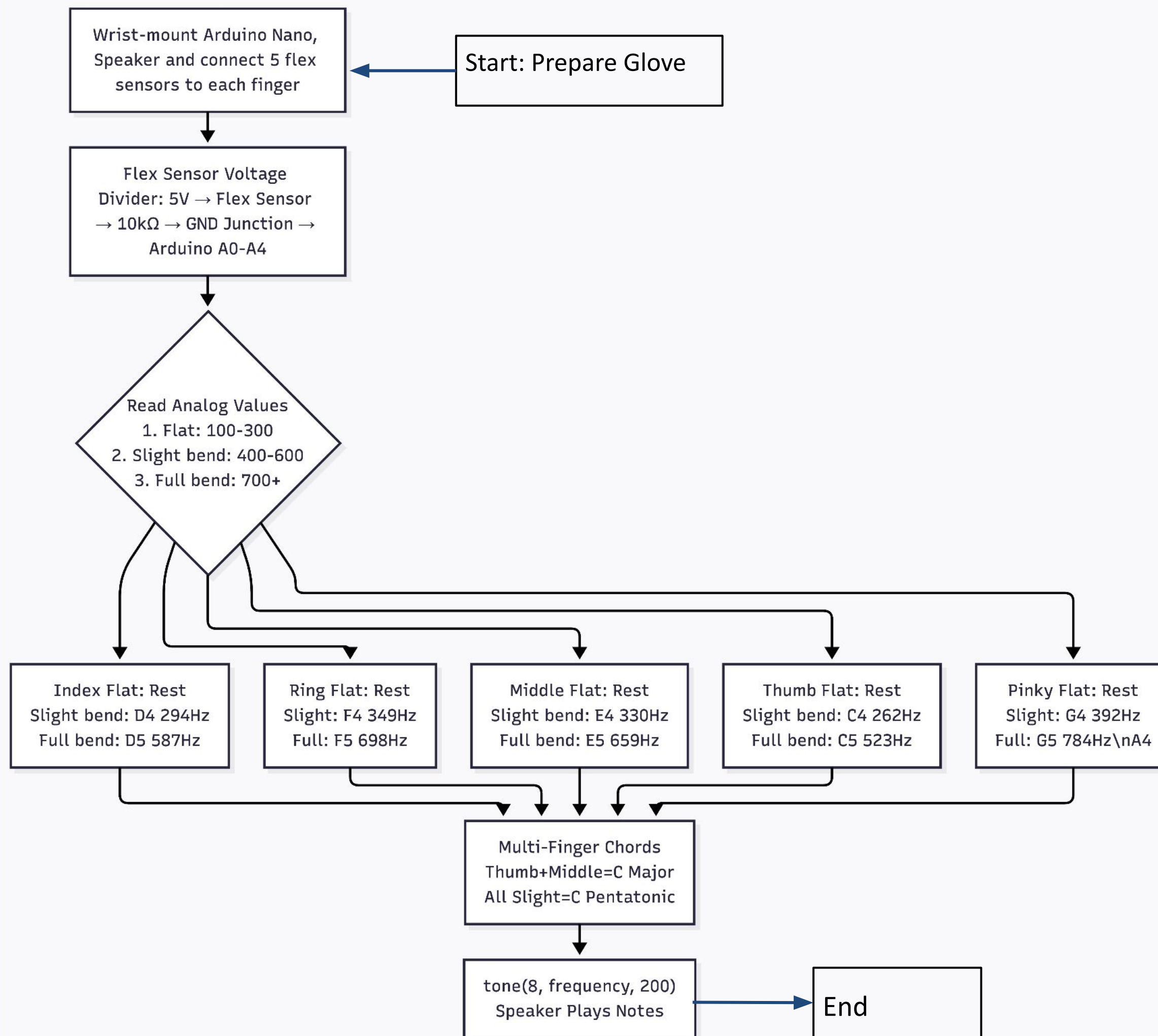
Signphony – A fusion of sign and symphony

We often gesture with our hands and jump around in joy, as if music were playing in the background. To us in the daily, these would be meaningless gestures. What if we set a new perspective to them, turning these supposedly random hand gestures into musical notes!

‘Maaya’ in Sanskrit means Magic.

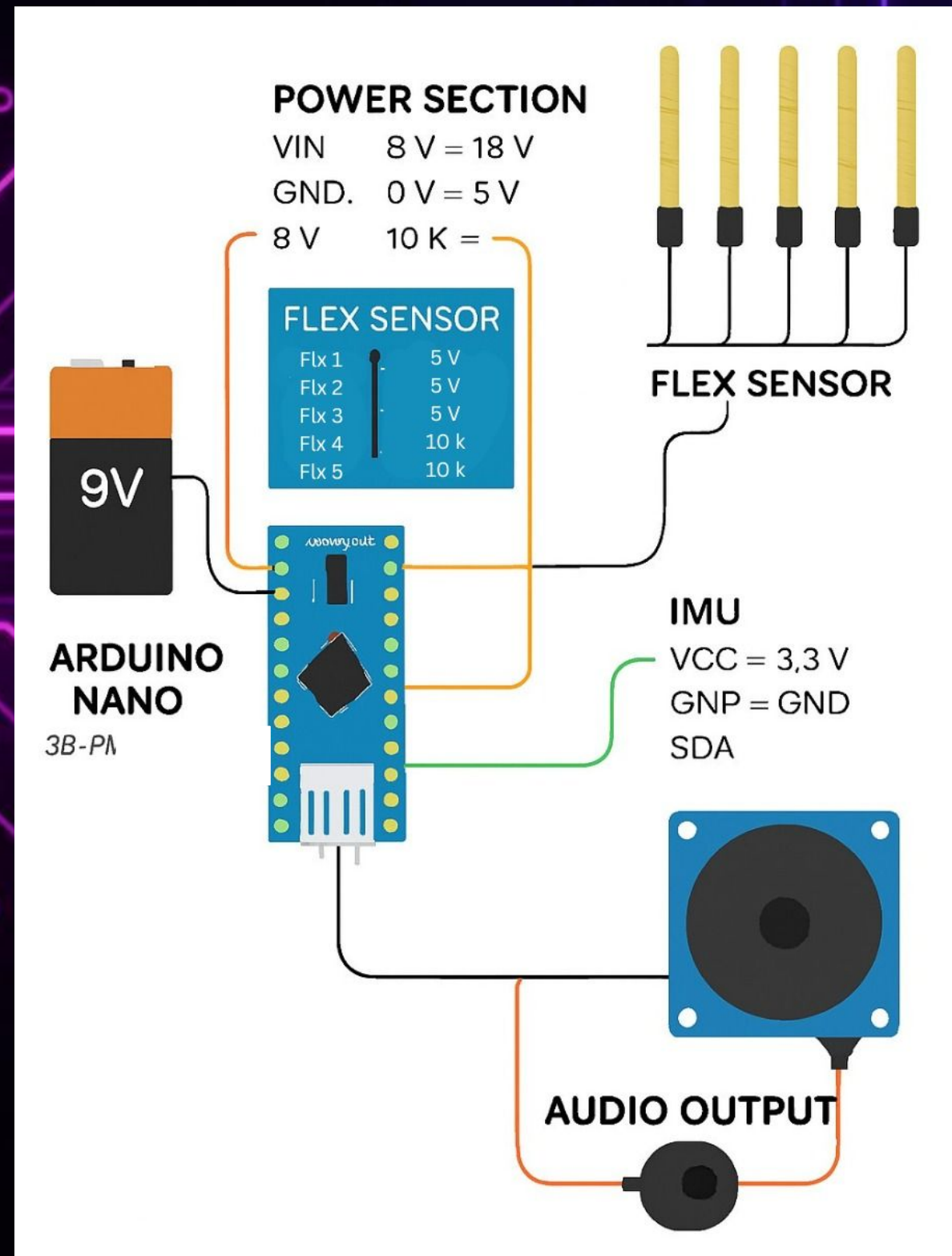
We use a simple glove, flex sensors, speakers ,arduino code and more to make this Maaya (Magic) come to life.

TATTVA 8 : Maaya - Perspective determines reality



METHODOLOGY

CIRCUIT DIAGRAM



WORKING IDEA

- Powered by an Arduino Nano, the glove reads sensor data and generates sound through a buzzer or speaker.
- Each finger and the amount of bending corresponds to a musical note.
- Signing a sentence creates a melody like playing music with your hands.
- It uses flex sensors and an IMU to detect motion, powered by an Arduino Nano.

COMPONENTS

COMPONENTS	AMOUNT
Arduino Nano Solderless	1
Flex Sensor 2.2	5
MPU-6050 IMU	1
Seeed Studio Grove Speaker	1
Jumper Cables Kit	1 (20-30 wires)
Breadboard Power Module 3.3V/5V	1
9V Battery	1
10K Ohm Resistors	5
Hot Glue Gun	1
ESD Glove	1

NOVELTY

Signphony, our Musical sign glove transforms sign language into live music, fusing communication and art. Each gesture is both a word and a note, each sentence both a message and a melody, making the human body itself the instrument.

Speak in signs, and the world hears a song.

For sign language users, everyday conversation becomes instant composition. For performers, it's a new instrument. For everyone, it's communication as art.

Tattva 8: Maaya - Perspective determines reality, states that our perspective determines what we experience as reality. The Musical sign glove reframes perception, where once we saw fingers signing words, and now we hear them composing music.



ELECTR VISTA

THANK YOU



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PCB CUPID